Rongxuan Zhou

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Research Interests

My research focuses on robot learning for contact-rich manipulation, in particular on world models that know the limits of their own perception and action. It follows three threads: latent world models that plan from vision and force, and the failure modes that arise when a sensing modality or an action direction is absent from the model; training-free safety and diagnostic methods that act on frozen checkpoints at sampling time; and the deployment of these models on real manipulators and humanoids.

Education

- **Northeastern University** *Sep 2024 – May 2026*
  M.S. in Robotics, Electrical and Computer Engineering — Advisor: Prof. Yang Ye Boston, MA
  - Thesis: *TACS: Training-Free Safety Guidance for Action-Space Diffusion Policies via Tweedie-Adaptive Constrained Sampling*
- **Northeastern University** *Sep 2020 – Jun 2024*
  B.E. in Mechanical Engineering and Automation Shenyang, China
  - Mechanical design lead, TDT Robot Innovation Laboratory; First Prize, 2021 RoboMaster University Championship

Research Experience

- **Northeastern University** *Feb 2025 – Present*
  Graduate Research Assistant — Advisor: Prof. Yang Ye Boston, MA
  Research on world models and safe policy learning for contact-rich manipulation. I lead two first-author lines (force fusion, action-space blind spots), co-develop the lab's other world-model projects, and own the lab's real-robot stack: controller, teleoperation, digital twin and data collection for the Franka FR3 and the humanoid experiments.
  - **ForceLeWM: A Modality-Robust JEPA World Model with Gated Force Fusion**
    - Built a JEPA world model fusing force/torque via a learned per-dimension gate: on simulated PushT-FT without force at test time, naive fusion cut planning success from 41% to 15%; gated fusion with alignment restored it to 35%.
    - Deployed one frozen sim-trained checkpoint per task on real Franka assembly, seating the part in 27 of 30 gear-meshing and 29 of 30 peg-insertion trials, with force from a proprioceptive wrench observer, not an F/T sensor.
  - **Subspace-Restricted Planning under Action Compression in Latent World Models**
    - Showed that action compression leaves latent-world-model planners partly blind: on a released Cube checkpoint 15 of 25 action dimensions never reach the predictor, so CEM moves randomly along them, unseen by model-side checks.
    - Proposed training-free pruned sampling from the frozen encoder's SVD, provably unable to worsen the best plan, plus an active-subspace cut on the cost-gradient spectrum: +10.7 success points on Cube, +12.3 on Cube-double.
  - **TACS: Training-Free Safety Guidance for Action-Space Diffusion Policies**
    - Added workspace safety to pretrained diffusion policies without retraining, by injecting barrier-function guidance into the denoising loop: 20× fewer constraint violations on robosuite Lift (6.6% → 0.32%, $p = 0.015$ over 5 seeds).
    - Applied the barrier correction to the Tweedie-predicted end-effector position at each denoising step, so later steps adapt the full action trajectory; on three robosuite tasks it beat post-hoc projection on the success–violation trade-off.
  - **PRISM: Prior-Guided Imagination Sampling in World Models**
    - Built the teleoperation and data stack on a Franka FR3, streaming Meta Quest 3 (WebXR) commands through a 1 kHz Cartesian impedance controller, and open-sourced a 500-demonstration PushT dataset collected with this stack.
    - Trained the JEPA latent world model (ViT encoder + action-conditioned autoregressive Transformer) on that dataset and deployed it closed-loop on the real robot with real-time MPPI sampling-based planning in the latent space.
  - **TOIF: Terrain-Aware Hierarchical Control with Proprioceptive Feedback for Humanoid Navigation**
    - Co-developed the feedback path of a three-tier humanoid navigation stack: proprioceptive anomaly detectors feed leaky integrators whose output becomes structured language for a planner that replans over a semantic scene graph.
    - Ran the Unitree G1 hardware validation on uneven, stepped and slippery terrain: transferred the Isaac Lab-trained anomaly detectors to the robot (AUC 0.87 and 0.94 on rough and stepped terrain) and closed the loop to replanning.
  - **EquiJEPA: Learning Action Representations for Effect Prediction and Action Generation**
    - Co-developed the state-blind effect head of EquiJEPA, which makes the same action yield one shared latent transition across states, raising its transition concentration from 0.747 to 0.910 on Push-T and from 0.570 to 0.872 on Cube.
    - Evaluated the learned action prototypes by physical execution, with cross-state reuse and unseen action composition: agent error at new Push-T states fell from 4.02 to 2.67 px and two-dot composition success rose from 11.5% to 40.1%.
  - **What Reveals Below: Fold-Contact Affordances for Cloth Unfolding**
    - Co-developed fold-contact affordances for dual-arm cloth unfolding: a transparent table gives a top mask and a bottom contact mask, made into fold residual, contact overlap and frontier maps of where hidden folds meet support.
    - Built the physical transparent-table dual-arm rig with two RealSense D435 cameras in top and bottom views, and ran the real-robot unfolding trials with the garment state segmented by SAM 3 from the synchronized observations.

- **Shanghai Jiao Tong University, IWIN-FINS Lab** *Jun 2022 – Aug 2024*
  Research Assistant — Advisor: Prof. Jianping He Shanghai, China
  - **FINES: Full-Stack Mobile Manipulation Platform — Hardware-Software Co-Design**
    - Architected the control stack from bare metal up (Master MCU, Intel NUC, watchdog) with a C++ embedded framework (FineMote) managing CAN/RS485/I2C at a 1 ms cycle, supporting 3 downstream research projects.
    - Designed a custom 6-DOF manipulator: actuator and transmission sizing, rigid-body dynamics, closed-form IK (Pieper) and MoveIt integration, closing the loop from CAD to real-time joint-space control on the embedded stack.
    - Built the perception pipeline end-to-end: multi-camera calibration, Point-LIO LiDAR-inertial odometry fused with UWB positioning, and YOLO + ICP obstacle detection.
    - Extended the platform to multi-agent coordination: synced the manipulator with the AGV over ROS 2 for mobile manipulation, and built an air-ground transport system with MPC quadrotor control validated hardware-in-the-loop.
- **Shanghai University, Key Lab of Intelligent Manufacturing and Robotics** *Jun 2022 – Apr 2023*
  Research Assistant — Advisor: Prof. Guiqin Li Shanghai, China
  - Developed an axisymmetric ABAQUS contact model of swage autofrettage in a 15-5PH steel high-pressure cylinder: a frictional mandrel contact pair, true stress-strain inputs and the Bauschinger effect predict the residual-stress field.
  - Swept mandrel interference from 0.5% to 2.5% in the parametric contact model to shape the compressive residual stress at the cylinder bore, finding that the compressed layer and the Bauschinger effect both grow with the interference.

Industry Experience

- **XtalPi** *May 2026 – Present*
  Robotics Software Engineer, Controls and Robot Learning — Full-time Cambridge, MA
  Built the control, simulation and learning stack of an autonomous chemistry lab delivered to Eli Lilly's US site, designed to run 24 station modules through up to 144 unattended reactions per order: experiment scheduling, arm control, vision-guided dosing, an LLM operations agent and module software.
  - Built the vision and simulation-learning loop for milligram powder dosing: a two-camera MobileNet-UNet measures each spatula tip's 6-DoF pose, and Blender scoop simulation trains an opt-in Rainbow DQN beside the rule-based loop.
  - Commissioned motion control for five CANopen arms: taught pose statemaps by three-point affine teaching, tuned PRM planning with link bounding boxes inside safe zones, and calibrated each joint's zero to a fixture in seven steps.
  - Built the experiment scheduler on a discrete-event model of the station: one simulation generates each order's process graph, scores ALNS plate layouts, returns per-plate ETAs and lets a follow-on order start before the last one ends.
  - Developed the LLM operations agent: 52 capabilities that validate designs, commit orders to the scheduler, run GO/NO-GO preflight and triage failures from logs, with three token-gated writes, 636 pytest cases, 109 behavior evals.
  - Developed module-level control software on the device middleware: Python ops over 17 driver types (eCAL and Cap'n Proto) for CAN, CANopen, Modbus and serial hardware, in a framework of 311 async ops with an E-stop state machine.
- **Dinnar Automation** *Apr 2025 – Dec 2025*
  Robotics & Machine Vision Engineer — Contract, Hybrid San Jose, CA
  Full-stack engineering contributor to the CSE Round Lighting Check & Final-Inspection (FI) AOI machine, a vision-guided robotic inspection system for Texas Instruments Chip Scale Element (CSE) products, built for 85K+ units/day.
  - Implemented the vision-guided inspection pipeline: an Epson SCARA robot, servo-driven reorientation and 4-angle hardware-triggered CCD inspection, covering 19 defect categories with 100% detection on customer-provided samples.
  - Wrote the classical-CV defect detectors in C++: NCC template matching with edge and blob checks for cracks, HSV thresholds for contamination, OCR with a sharpness measure for markings, and a dark-field test for light leakage.
  - Reached a 3.8 s takt per four-part batch by imaging the bottom face in transit under strobe light instead of at a full stop, and added an NG reconfirmation camera that re-images the flagged parts off the critical path to reduce false rejects.
- **CloudMinds Robotics** *Dec 2023 – Apr 2024*
  Robotics Control Engineer — Internship, On-site Shanghai, China
  Contributed real-time control infrastructure for multi-robot assembly cells used in harmonic-reducer production, plus a data-driven UR5e impedance controller, bridging factory-grade automation with manipulation research.
  - Rebuilt the manipulator's real-time motion-control communication on EtherCAT using IEC 61131-3 Structured Text, cutting the joint control cycle by 40% and delivering sub-millimeter trajectory repeatability as measured per ISO 9283.
  - Added a learned residual to the analytical Hogan impedance model on a UR5e for contact-rich assembly: a 42% vibration reduction in a 72-hour continuous aging test against the purely analytical baseline on the same assembly task.
  - Built an OPC UA bridge between ROS 2 and the plant's Siemens S7-1500 PLCs, so research controllers could run on the production PLC network at $\leq \pm 2$ ms jitter with no custom protocol on the PLC side and no changes to the plant stack.
- **NIO Automobile** *Jul 2023 – Oct 2023*
  Autonomous Driving Research Intern — Internship, On-site Shanghai, China
  Research internship on the autonomous-driving team: generative data augmentation for ECU testing and sim-to-real transfer for vehicle lateral control.
  - Built a GAN-based CAN-bus signal synthesizer to augment safety-critical edge cases, producing 5× rare-event training data, expanding ECU long-tail test coverage by 35% and surfacing failure modes that classical fuzzing had missed.
  - Domain-adversarial sim-to-real policy transfer (DANN): implemented gradient reversal layers that align CARLA and NIO ET5 feature distributions; in closed-loop road tests run by the vehicle team, lateral-control error fell by 37%.

Publications

C=Conference, J=Journal, S=In Submission, †=first author

[S.6] **R. Zhou†**, Y. Wang, X. Hu, and Y. Ye (2027). **ForceLeWM: A Modality-Robust JEPA World Model with Gated Force Fusion**. In submission, *IEEE International Conference on Robotics and Automation* (*ICRA*).

[S.5] **R. Zhou†**, Y. Wang, X. Hu, S. Cao, and Y. Ye (2027). **Subspace-Restricted Planning under Action Compression in Latent World Models**. In submission, *International Conference on Learning Representations* (*ICLR*).

[**S.4**] Y. Wang, J. Xia, B. Sun, **R. Zhou**, X. Hu, Y. Shi, J. Du, and Y. Ye (2027). **PRISM: PRior-guided Imagination Sampling in World Models**. arXiv preprint 2026; in submission, *IEEE International Conference on Robotics and Automation* (*ICRA*).

[**S.3**] Y. Wang, H. Zhang, Q. Yin, **R. Zhou**, X. Hu, Y. Shi, and Y. Ye (2027). **EquiJEPA: Learning Action Representations for Effect Prediction and Action Generation**. In submission, *International Conference on Learning Representations* (*ICLR*).

[**S.2**] **R. Zhou**† and Y. Ye (2026). **TACS: Training-Free Safety Guidance for Action-Space Diffusion Policies via Tweedie-Adaptive Constrained Sampling**. In submission, *IEEE Robotics and Automation Letters* (*RA-L*). Master's thesis.

[**S.1**] X. Hu, L. Sun, Y. F. Yeung, Y. Wang, **R. Zhou**, and Y. Ye (2026). **What Reveals Below: Fold-Contact Affordances for Cloth Unfolding**. In submission, *Conference on Robot Learning* (*CoRL*).

[**C.2**] Y. Wang, **R. Zhou**, C. Wang, X. Li, Z. Teng, X. Hu, A. Ramezani, J. Du, and Y. Ye (2026). **Thinking on Its Feet: Terrain-Aware Hierarchical Control with Proprioceptive Feedback for Humanoid Navigation**. In *IEEE-RAS International Conference on Humanoid Robots* (*Humanoids*).

[**C.1**] W. Xi, J. Guo, C. Wang, S. Wu, and J. He (2025). **A Robust Distributed Odometry for Mobile Robots with Steerable Wheels**. In *IEEE/RSJ International Conference on Intelligent Robots and Systems* (*IROS*). Contribution: robot structural and simulation developer.

[**J.1**] L. Sun, **R. Zhou**, G. Li, J. Li, and P. Mitrouchev (2023). **Simulation Research on Residual Stress of Swage Autofrettage-processed High-Pressure Cylinder**. *Journal of Physics: Conference Series*, Vol. 2587, 012088 (ICAMMT-2023).

Open-Source Software

- **frlink — real-time control and teleoperation stack for Franka FR3** *2026*
  C++ / Python, libfranka @Rongxuan-Zhou/frlink
  - 1 kHz torque-level Cartesian impedance controller on an isolated-CPU real-time host, with a client side for Meta Quest 3 (WebXR) teleoperation and policy action chunks; 0.10 mm drift and zero missed cycles in a 5-minute position hold.
- **MuJoCo MCP Server** *2025*
  Python, MuJoCo, Model Context Protocol @Rongxuan-Zhou/mujoco-mcp-server
  - A 65-tool Model Context Protocol server that lets LLM agents drive MuJoCo simulation: loading of 50+ Menagerie robots, inverse kinematics, trajectory optimization (iLQR and MPPI), domain randomization and RL environments.

Skills

- **Robot Learning:** Diffusion Policy, World Models (JEPA), Imitation Learning, Reinforcement Learning (Rainbow DQN), VLA, sampling-based planning (MPPI, CEM), LLM agents (tool use)
- **Real-Robot Systems:** Franka FR3, UR5e, Unitree G1, teleoperation (WebXR, Meta Quest 3), sim-to-real, force/torque sensing, ROS 2, CANopen, EtherCAT, eCAL, MoveIt, FANUC, ABB
- **Control:** 1 kHz torque control, Cartesian impedance control, kinematics (FK / IK), rigid-body dynamics, trajectory tracking, PRM motion planning, PLC programming (IEC 61131-3)
- **Perception & Vision:** Multi-camera and hand-eye calibration, LiDAR-inertial odometry (Point-LIO), YOLO, U-Net segmentation, SAM, ICP, OpenCV, AOI system design
- **Simulation:** MuJoCo (MjSpec), Isaac Sim / Isaac Lab (PhysX), Genesis, Newton, pymunk, Blender, MJCF, digital twins, domain randomization, discrete-event simulation (SimPy)
- **Hardware & Embedded:** EtherCAT, CANopen drives (CiA 402), STM32, Jetson, SolidWorks, ANSYS, ABAQUS, CODESYS, Siemens PLC, OPC UA, CAN / RS485 / I2C
- **Programming & Infra:** Python, PyTorch, CUDA, C/C++, MATLAB, Linux, Git, Docker, CI/CD, Hugging Face datasets